EECS 767
Information Retrieval

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Information retrieval

What is information retrieval (IR)?

Information retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need from within large collections (usually stored on computers).

- Introduction to information retrieval

EECS767: Topics

- Text IR
  - Boolean model
  - Vector Model
  - Text algorithms
  - Evaluation
- Web IR
  - Web crawling
  - Link analysis, Page Rank
- Emerging Topics
  - Machine learning in IR
  - Social network analysis

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- OK, are you still interested in the class?

Who's who?

- Instructor: Bo Luo
  - Professor at KU EECS
  - Ph.D. Penn State 2008
  - Research interests:
    - Adversarial/trustworthy ML
    - IoT Security
    - Information security and privacy
    - Security and privacy in the context of Internet and information retrieval
  - Contact me:
    - bluo@ku.edu
    - http://www.ittc.ku.edu/~bluo/
    - **By appointment**

Time and location

- Class: M 6:00pm – 8:50pm,
- LEEP2 2300

- Office hours: Thursdays 2pm to 3pm @ Eaton 2016
- Or by appointment

EECS767: Textbook

- Textbook:
  - *Introduction to Information Retrieval*, by Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze, Cambridge University Press. 2008.
  - **Available online**
- For reference
  - *Modern Information Retrieval*, by Ricardo Baeza-Yates and Berthier Ribeiro-Neto, ACM Press, 1999.
  - *Information Retrieval: Data Structures and Algorithms*, by William B. Frakes and Ricardo Baeza-Yates, Prentice Hall, 1992.
  - *The Geometry of Information Retrieval*, by C. J. van Rijsbergen, Cambridge University Press, 2004.

EECS767: Schedule

- We will use Canvas
  - http://canvas.ku.edu/
  - Slides will be posted (usually before class)
  - Project reports should be submitted through Canvas.
- Make sure you check emails from Canvas!
- Check email before you come to class.

EECS767: Tasks and Grading

- Course Project (20%)
- Exams (75%)
  - Two exams: 40%+35%
- In-class Quizzes (5%)
  - In-person, in-class quizzes, 1% each
  - No grading, answers will be provided
  - You get full credit as long as you submit

Exams

- Two exams
- Tentative dates
  - Exam 1 (40%): 03/09
  - Exam 2 (35%): 04/20
- I reserve the right to adjust exam dates.
- Schedule changes will be announced in class and updated on Canvas.

- **Exams are NOT cumulative**

- **No exam during the finals' week.**

EECS767: Grading

- We will use plus/minus grading
  - A: 93+
  - A-: 90+
  - B+: 87+
  - B: 83+
  - B-: 80+
  - C+: 77+
  - C: 73+
  - C-: 70+
  - D: 60+
  - F: 60-

Policies: Academic Integrity

- Academic Integrity!

  *Academic integrity is a central value in higher education. It rests on two principles: first, that academic work is represented truthfully as to its source and its accuracy, and second, that academic results are obtained by fair and authorized means. "Academic misconduct" occurs when these values are not respected. Academic misconduct at KU is defined in the University Senate Rules and Regulations. A good rule of thumb is "**if you have to ask if this is cheating, it probably is.**"*

  -- Office of the Vice Provost for Student Affairs.

  This is Serious!

Policies: Academic Integrity

- Your work will be manually graded and checked with plagiarism detection software.

- Make proper citations!!
- Rule of thumb: make it clear (to the grader) between your contribution and the literature.

- If you are unsure, it's probably NOT OK!
- If you are unsure, ask before you do it!

Policies: Attendance

- Attending class is one of the most reliable ways to stay current and perform well.
- I will make a good-faith effort to post important materials to Canvas (e.g., slides, announcements, etc).
- I cannot guarantee that every item will be posted immediately or with full details. If you miss an announcement (e.g., deadline reminders, exam logistics, or schedule changes) because you were absent, you remain responsible for knowing and following course requirements.
- There will be five in-class quizzes during the semester. They are not graded. You get full credit as long as you submit and reasonably answer every question (even if your answers are incorrect).

Policies: Exams

- Objective: to provide a **fair, consistent, and accurate** evaluation for every student.
  - **Pre-approved absence (advance notice required).**
    - Only if: (1) You have a valid reason to miss class and (2) you obtain my email approval before the exam begins (typically well in advance, not minutes before).
    - Bad (unacceptable) examples: Attending friends' weddings or social events; Family vacations or travel plans; Leaving early for breaks or returning late from breaks; Transportation difficulties (car trouble, missed bus); Being busy with another course; Oversleeping, forgetting the exam, misreading the schedule, or calendar mistakes.
    - Valid reasons: official university obligations (e.g., approved athletic travel), court/legal obligations, attending conferences to present papers (need note from advisor), other circumstances that are both **serious** & **unavoidable**.

Policies: Exams

- Objective: to provide a **fair, consistent, and accurate** evaluation for every student.
  - **Documented emergency (no advance notice possible).**
    - Genuine emergency, **verifiable documentation** required.
    - Good examples: A doctor's note for significant illness/injury requiring urgent care; A police report for a car accident; etc.
    - Bad example: my father/mother/friend is a doctor, who told me to drink a lot of water and sleep.
    - Please: Contact me as soon as you reasonably can after the emergency, and provide documentation sufficient for me to verify the circumstance.
    - If documentation is incomplete or does not substantiate the emergency, the request may be denied to preserve fairness to all students.

Policies: Exams

- Five quizzes, similar to exams; are easier and have fewer questions; count less toward the final grade.
- The same exam policies apply to quizzes.
- If you miss class or arrive late such that you miss the quiz submission or the attendance sign-in, you will receive a 0 for that quiz, unless your absence is formally excused under the Exam Attendance Policy.

Policies: Exams

- Additional notes to ensure fairness
  - Make-up exams may be scheduled at a different time and may be a **different version**. It will cover the **same learning objectives** and be of **comparable difficulty**, but not identical.
  - If you anticipate a potential conflict, contact me **as early as possible. Last-minute requests are rarely feasible.**
  - Students will be required to **spread out** and maintain a **reasonable distance** during exams. I and/or the TA may ask students to **change seats or assign seats to students prior to the exam**. Students are expected to comply promptly and respectfully.

EdTech Services

- KU School of Engineering is aware that some students are actively posting/downloading homework, lab, exam questions/responses to EdTech services (e.g., Chegg) even during exam time frames.
- The "terms of service" that are agreed to **do not protect the person when KU and/or the School of Engineering decide to conduct investigations related to academic misconduct (e.g., plagiarism and/or cheating)**.
- EdTech services,retain contact information of students who use their services and will release that information, which is traceable, upon request. Using these services for cheating purposes constitutes academic misconduct, which is **not tolerated in the School of Engineering**. It violates Article 3r, Section 6 of its Rules & Regulations, and may lead to **grades of F in compromised course(s), transcript citations of academic misconduct, and suspension (even expulsion) from the University of Kansas.**

Discussion and collaboration

- You are absolutely encouraged to discuss with your classmates about your homework assignments and projects.
  - **Each student should write down his/her own solution**.
  - Team competition: no discussion!
  - You are responsible for all your works
- For homework assignments and projects resulted from discussion, please always acknowledge other people's contribution by including a sentence at the beginning of the hand-in saying "I discussed the homework (project) with XXX (include more names if necessary)". There is absolutely NO penalty for doing so.

Generative AI

- Generative AI is useless if you don't know how to evaluate and adapt what it produces.
- You may use ChatGPT and other generative AI tools in this class in some instances: generating ideas, outlining steps in a project, finding sources (need to double check because ChatGPT produces phantom sources), getting feedback on your writing, and overcoming obstacles in projects.
- **Turning in work you have cut and pasted from generative AI is academic misconduct. It's no different from cutting and pasting from a website.** It is also academic misconduct if you use AI tools when you are asked not to. Read the university's policy on academic misconduct and the KU Code of Ethical Conduct and follow those in all your work.

Accommodations for Disabilities

- The Student Access Center coordinates academic accommodations and services for all eligible KU students with disabilities.
- If you have a disability for which you wish to request accommodations and have not contacted SAC, please do so as soon as possible. They are located in **22 Strong Hall** and can be reached at **785-864-4064 (V/TTY).** Information about their services can be found at www.access.ku.edu. Please contact me privately in regard to your needs in this course.

EECS767: Policies

- Grade dispute: within one week of returned grade.
- Late policy
  - Presentation: no late presentation accepted
  - Project reports: 10% penalty per day
  - Exceptions
    - Before deadline: instructor's approval.
    - After deadline: emergency only.

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- Any Questions?